Driving AI forward through Advanced Packaging Creativity

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ASE, Inc.

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aseglobal.com

AI IS HERE



HEALTHCARE



TELECOMMUNICATIONS



RETAIL

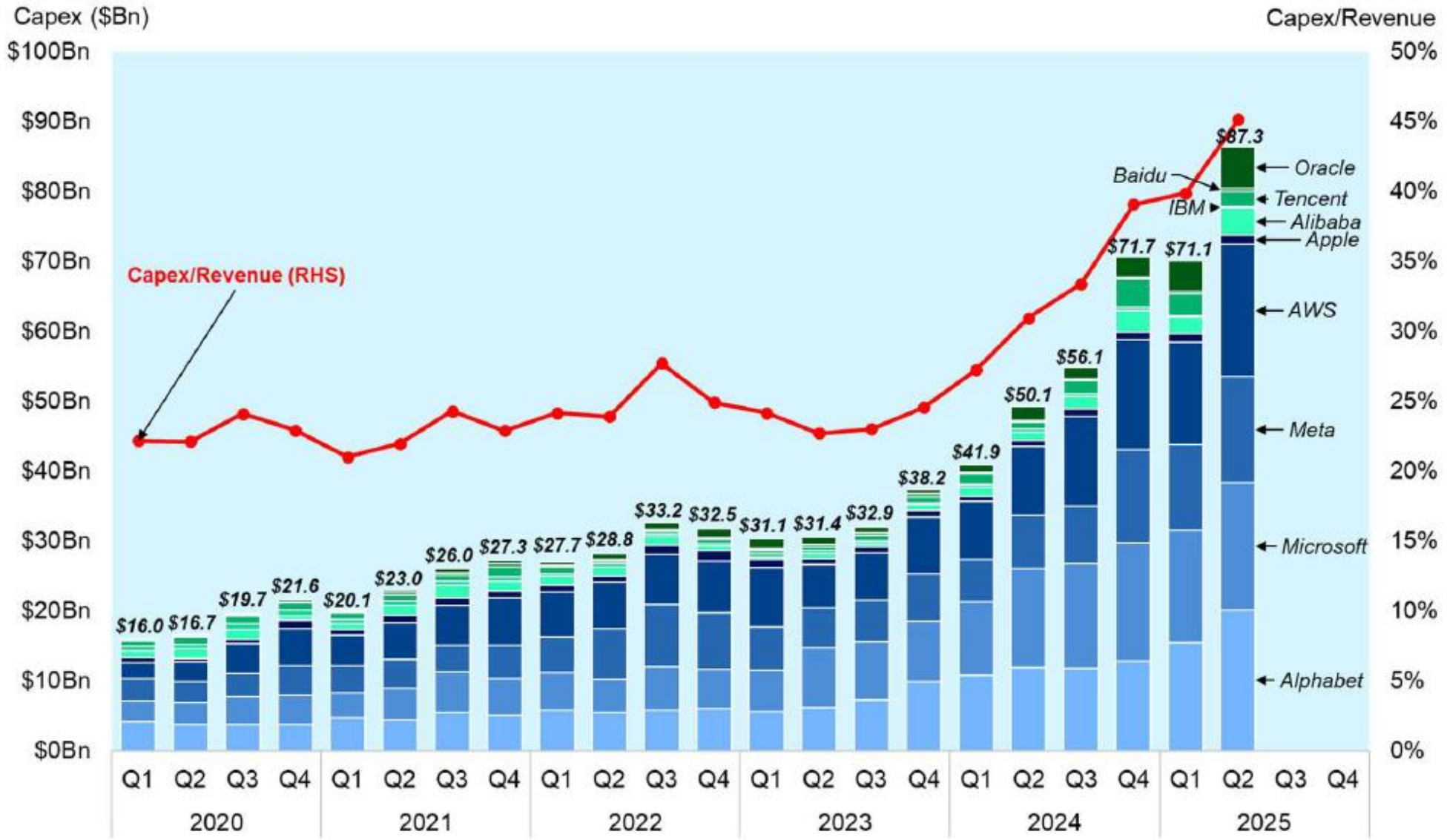


FINANCIAL SERVICES

- AI applications permeating global life, creating new efficiencies and new markets, from data center to edge devices.
- AI economy projected to soar from $189 billion in 2023 to $4.8 trillion by 2033 – a 25-fold increase in just a decade
- AI adoption could boost global GDP by 15% by 2035, based on game-changing capabilities and safe deployment.
- Unprecedented data generation is driving global datasphere trajectory towards 200ZB by 2030

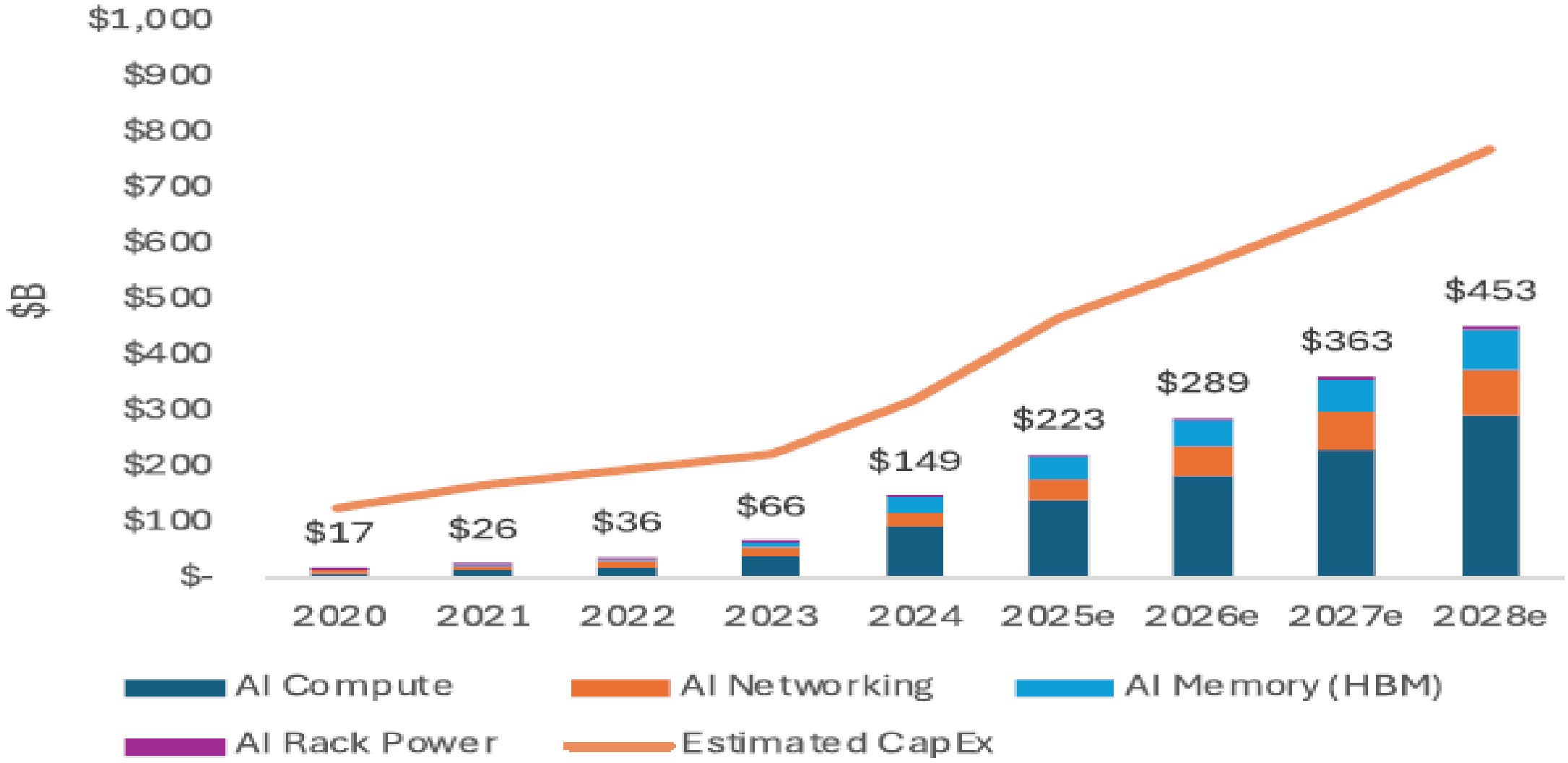
Source:
UN Trade and Development (UNCTAD), PwC, IDC 2025

Explosive Growth in AI Spending



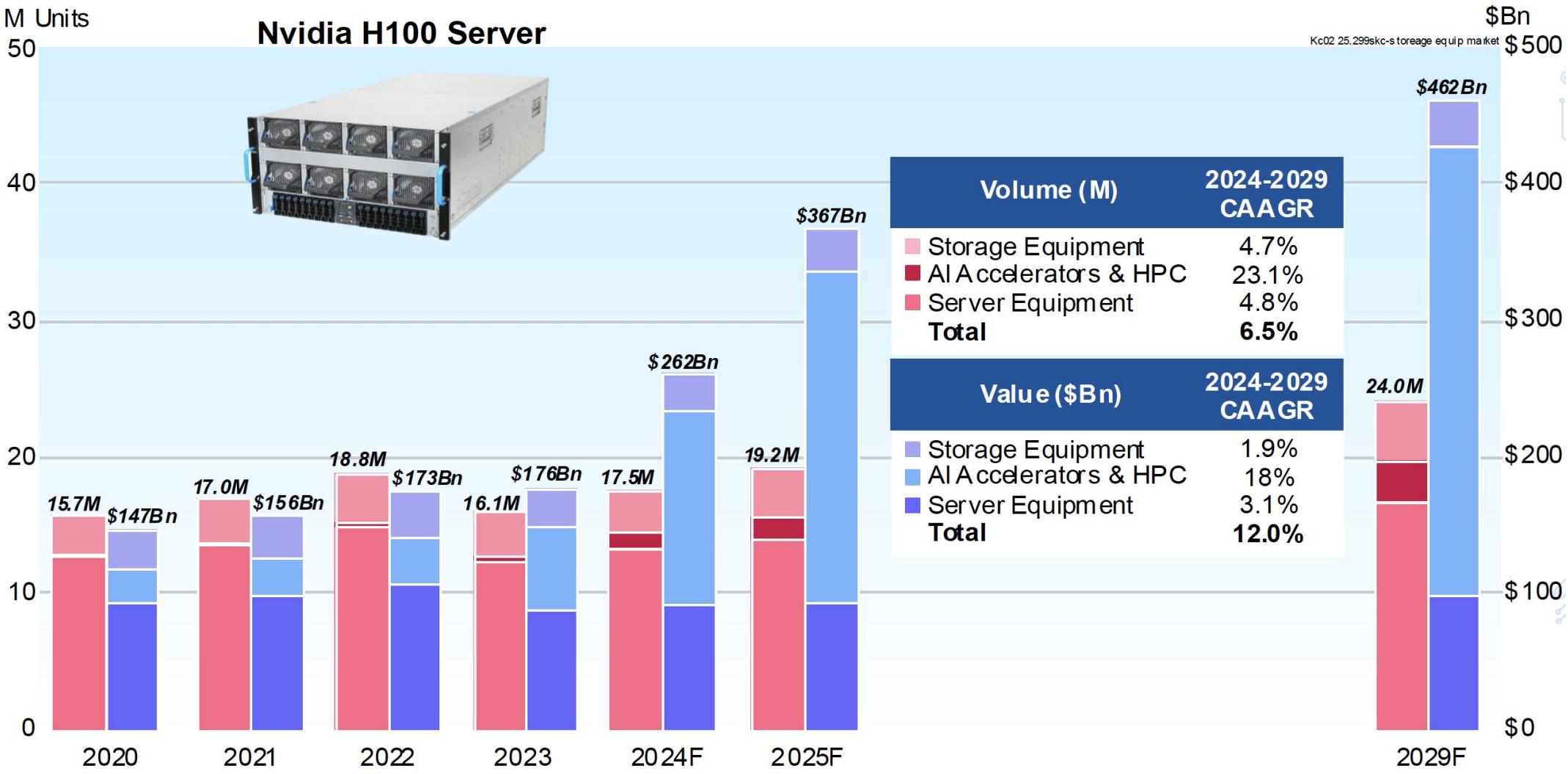
Source: Prismark

Data Center Capex & AI Semiconductors



Source :IDC

AI Equipment Market



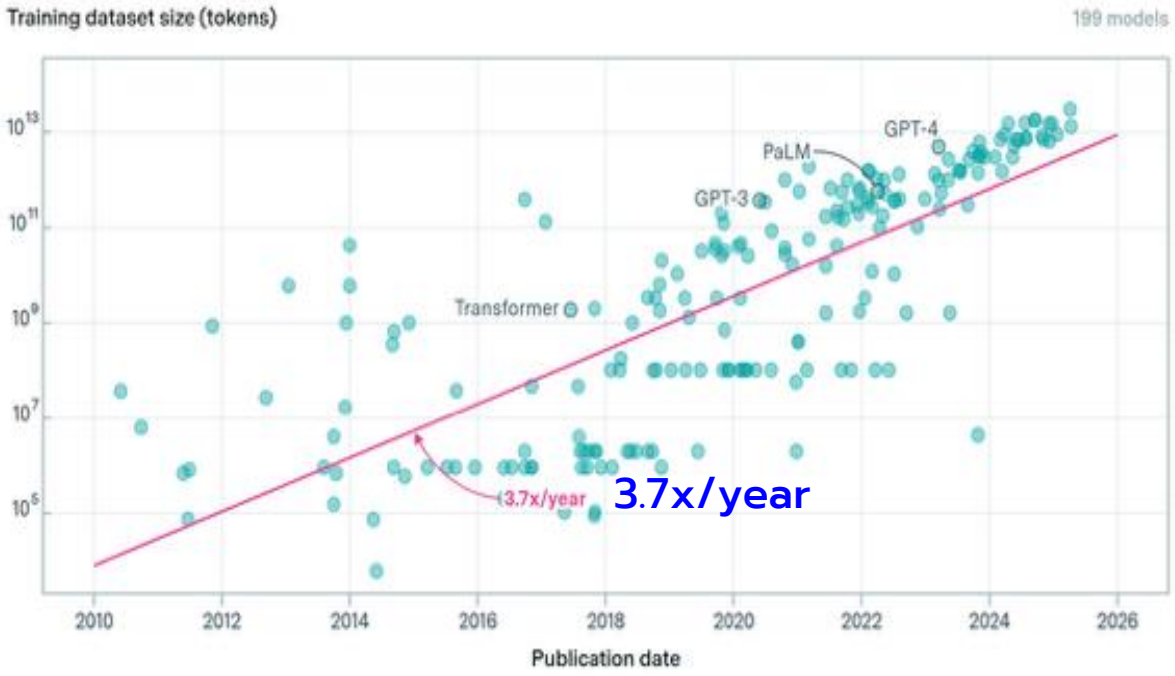
Growing Demands & Challenges in AI Era

Performance	Memory	Area	Power	Thermal
2x - 7x increased compute workload	**2x – 10x** network bandwidth & memory capacity	**2x – 12x** reticle **2x – 6x** pkg size (120x180 mm^2)	Exascale **21 MW** to zettascale **500 MW** power delivery	**~3kW** thermal cooling

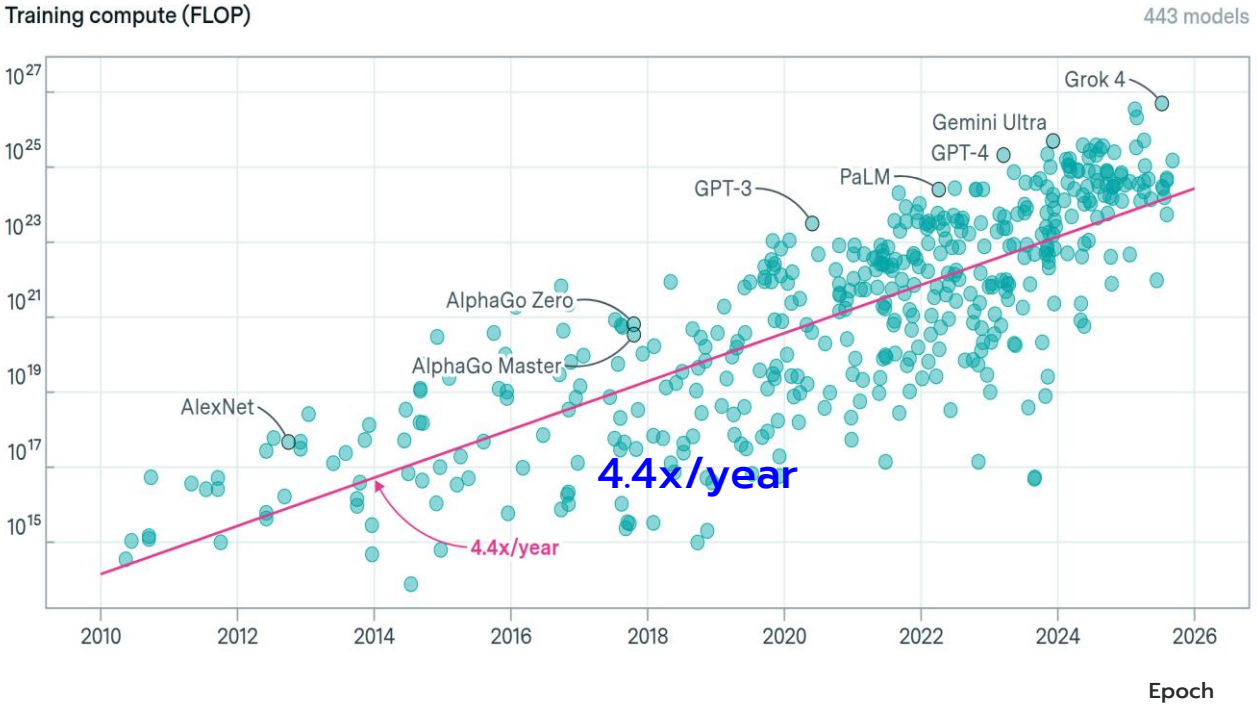
High Performance Demands by AI Industry

- Across all domains of AI, dataset size for train models doubles every 6 months with tens of trillions of words
- High-performance compute to train models grew 4-5x yearly 2010~2026 with 1.7x/year increase chip quantity & 1.4x/year improvement in performance per chip

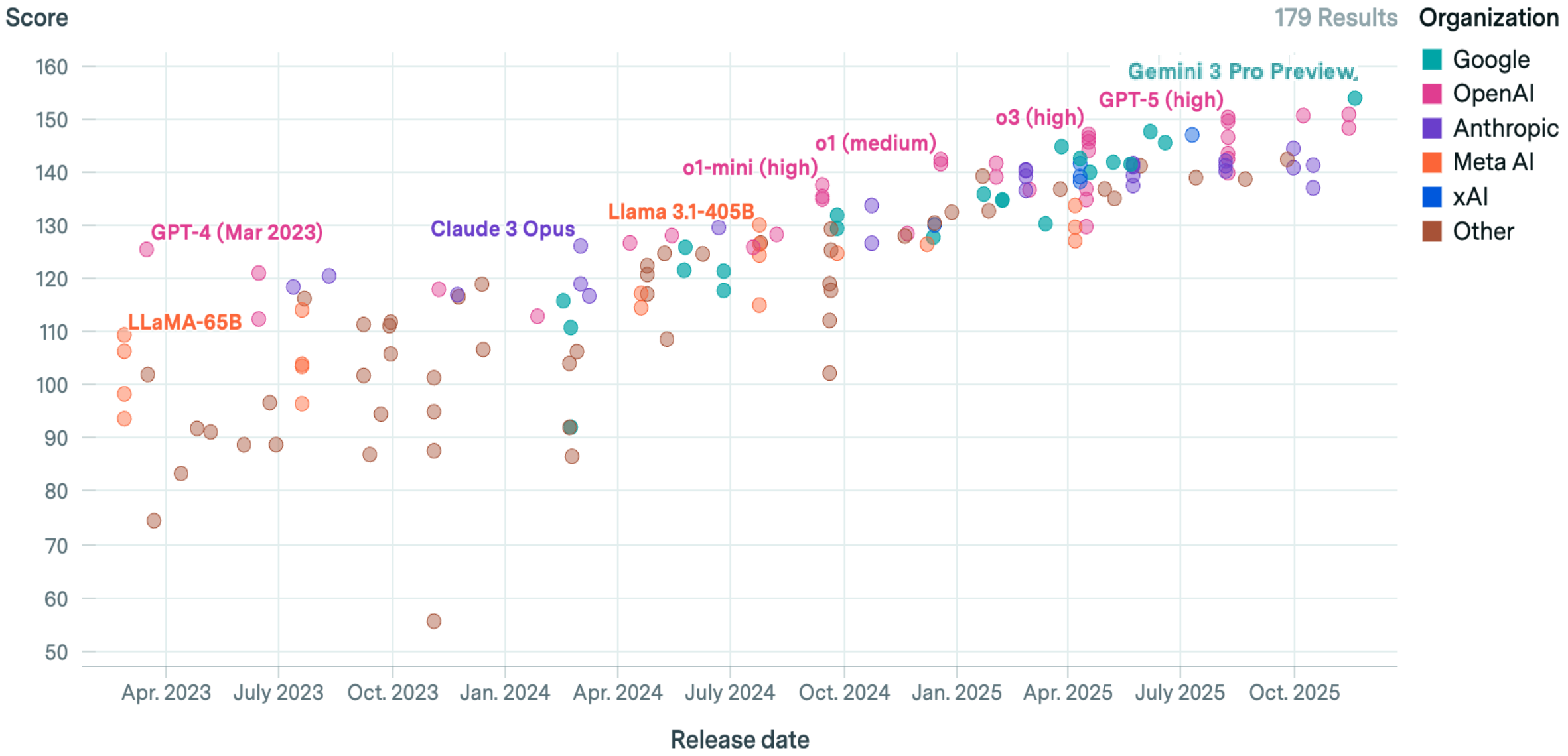
Size of Dataset



Super Compute Performance



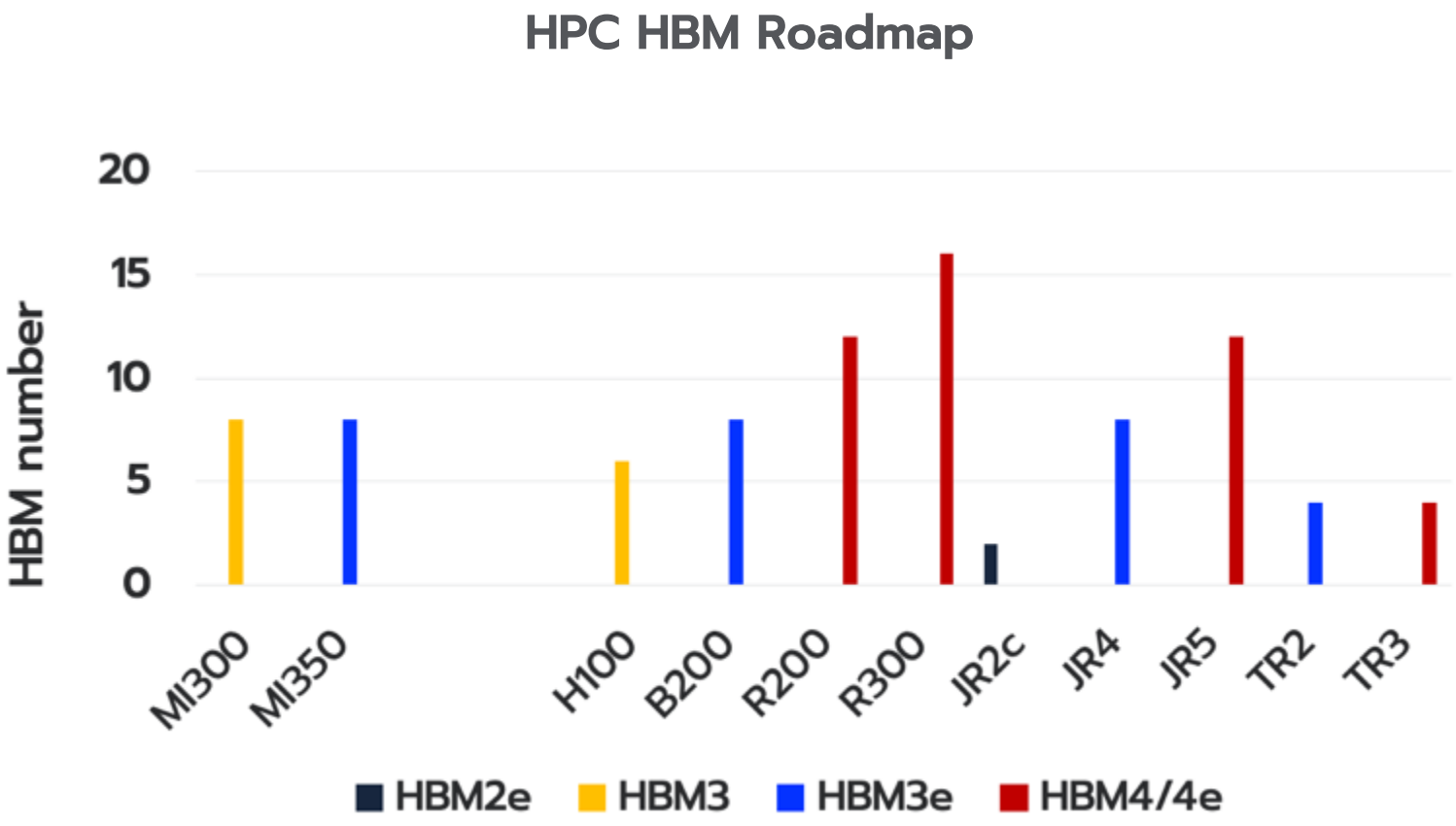
Performance of Leading AI Models

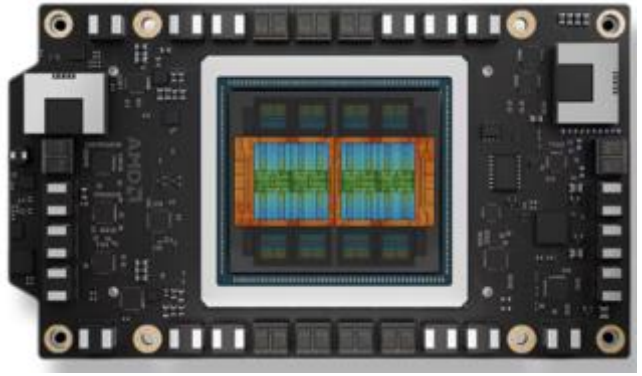


Source Epoch AI

HPC HBM Roadmap

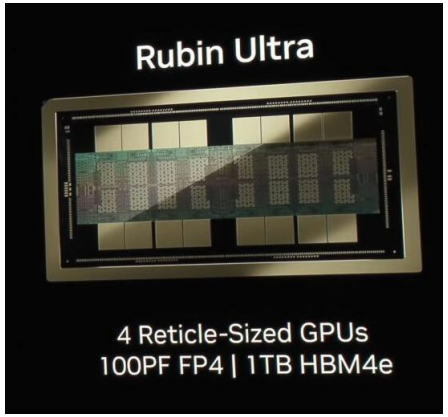
- HBM3e to HBM4/4e will increase from 8 to 12 to 16 by 2027~2028
- Larger module size required for greater HBM integration





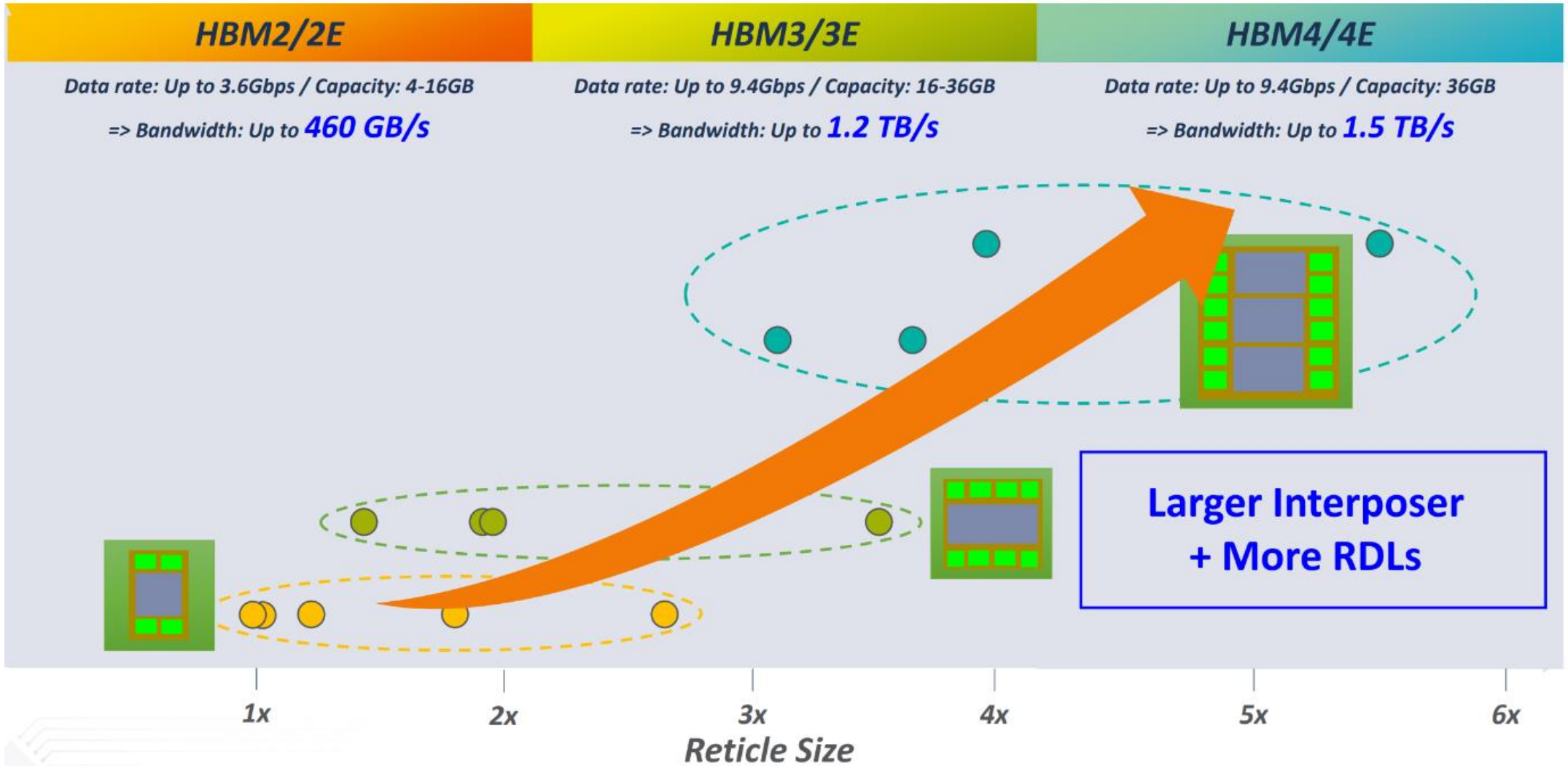
AMD MI350 (N3/N6, 18 chiplets, SOIC + CoWoS-S; 8HBM3e 288GB)

Source AMD



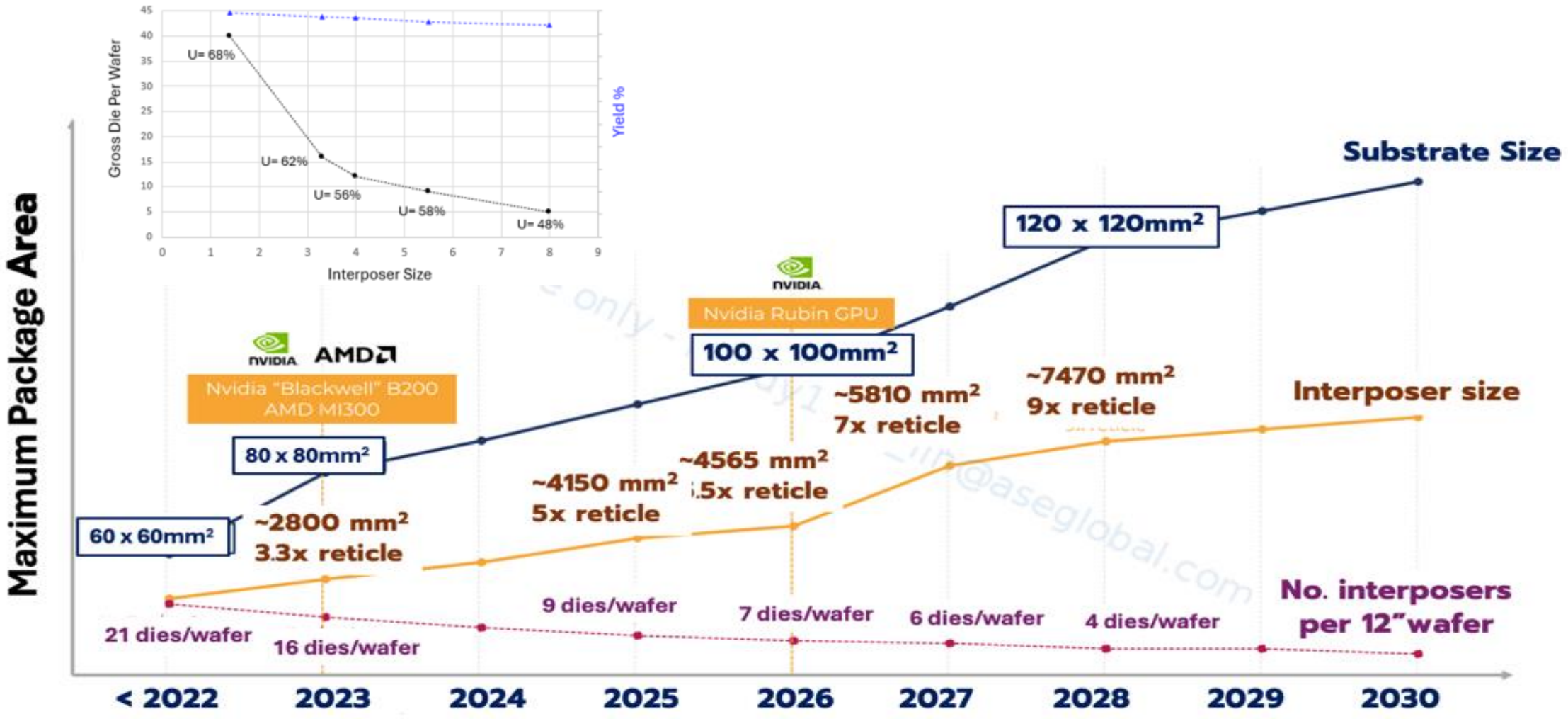
Nvidia Rubin Ultra CoWoS-L 16HBM4E,
Interposer size 124x50mm
Substrate size 153x77.5mm Source Nvidia

HPC & HBM Integration Trend



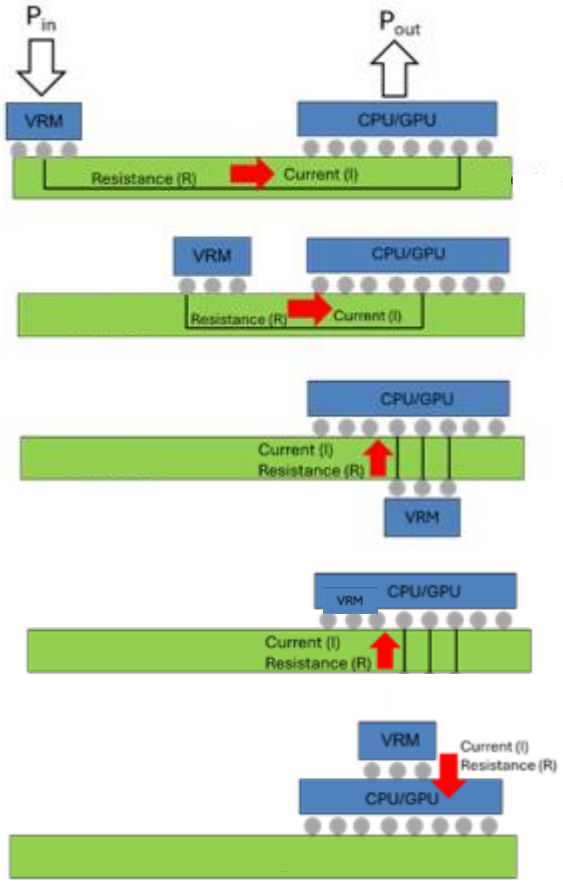
Key Challenges for Large Sized Modules

- Yield and utilization decreases when module size increases



Yole

Power Delivery Challenges



MBVR – Lateral PD

MBVR – Vertical PD

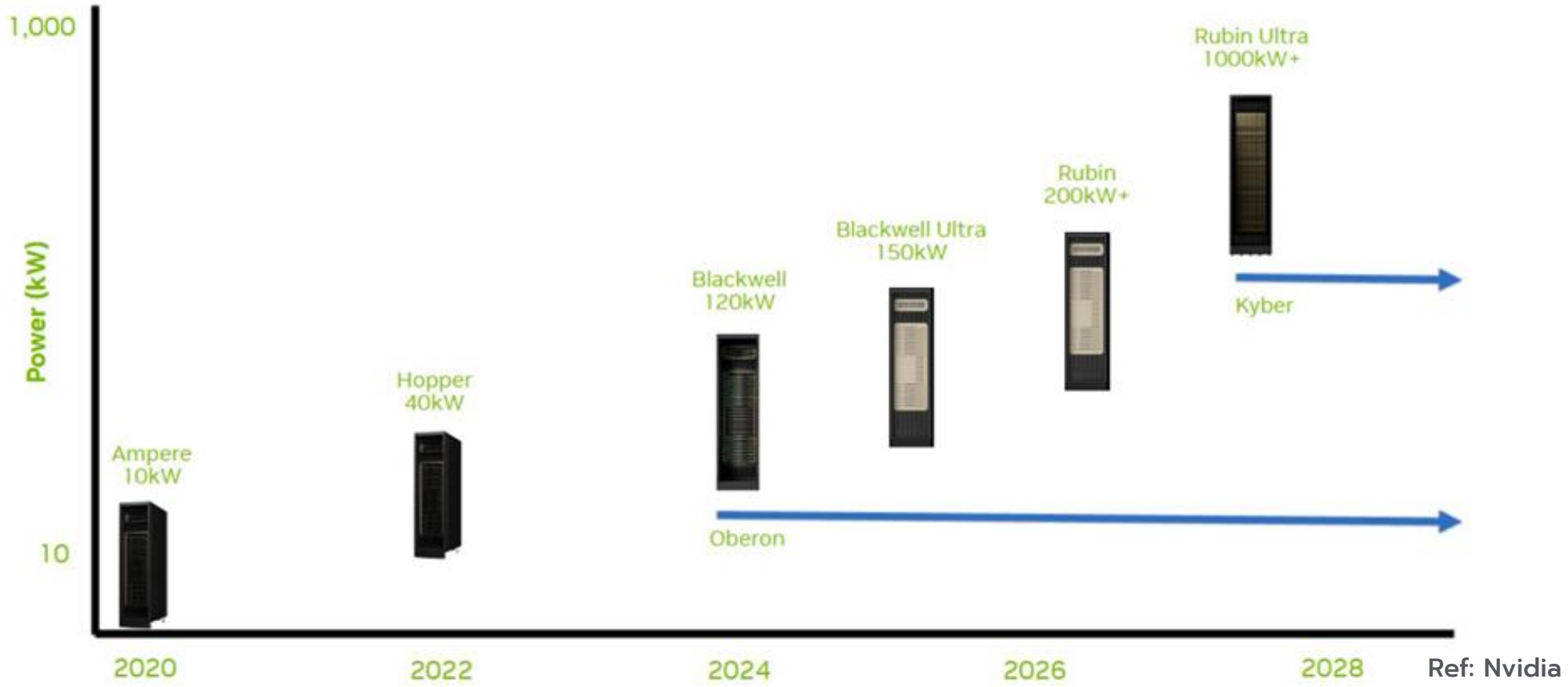
IVR Options
- Substrate, RDL
- Low, High Voltage

Backside PD

- PCB, Package Resistance → Routing losses
  - Distance of VRM to Load, voltage levels
  - Doubling of current, results in 4X power loss (I^2R)
  - Package power can be >10% of total TDP, with large IR drops
- Lower operating voltage and increasing voltage domains → higher load currents
- PDN impedance (Z) over frequency
  - Z needs to decrease in proportion to maintain acceptable voltage droop
  - Under package capacitor banks seeing diminishing returns in lowering Z to match current step/speed increases
- High phase count external VRM is spatially inefficient

MBVR – Motherboard Voltage Regulator; PD – Power Delivery; IVR – Integrated Voltage Regulator; RDL – Redistribution Layer

Ai Compute Rack Power Roadmap

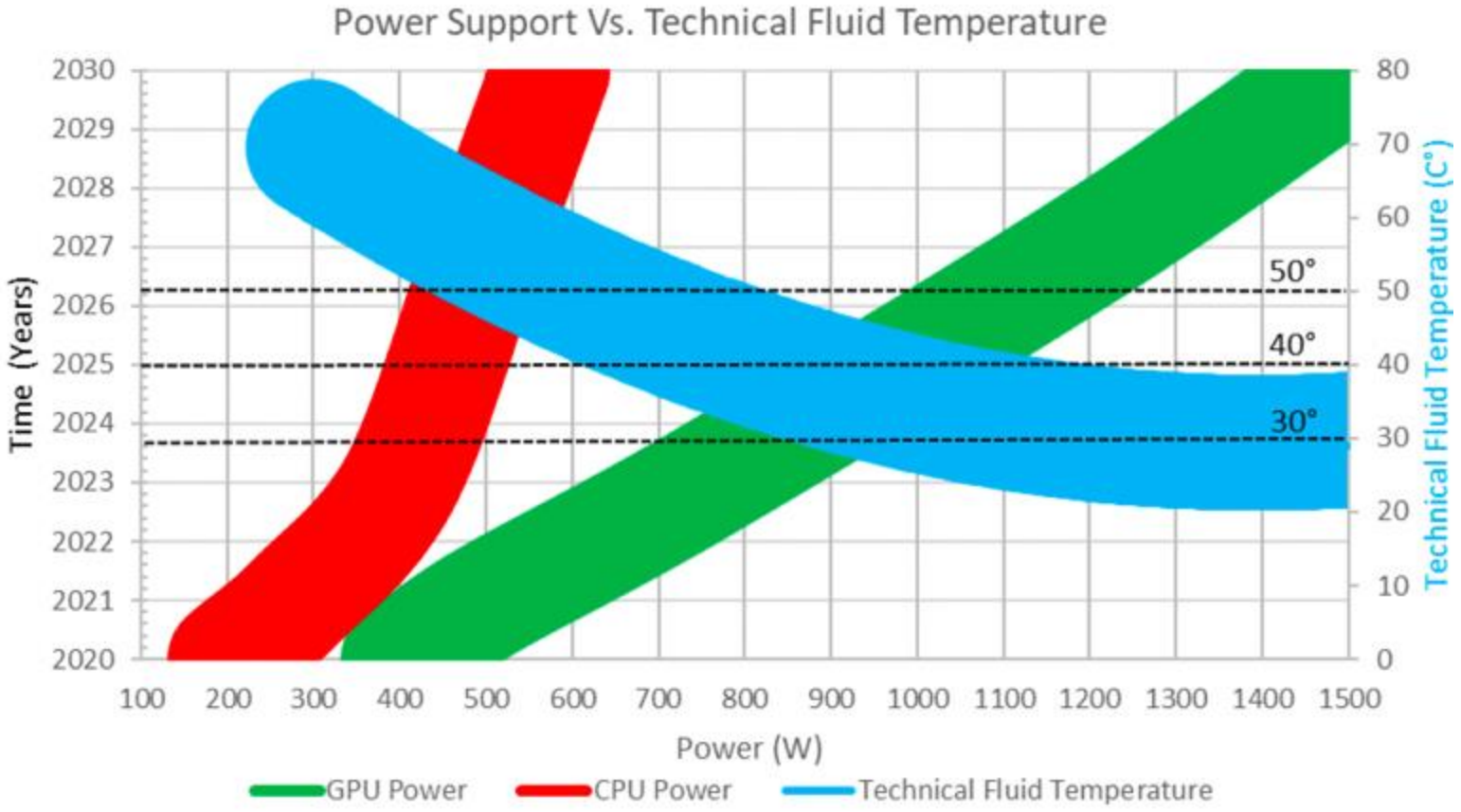


Ref: Nvidia

- AI workloads driving rack wattage from KW to MW
- Fundamental shift in power infrastructure to support the scale and power demands

Thermal Requirements for AI

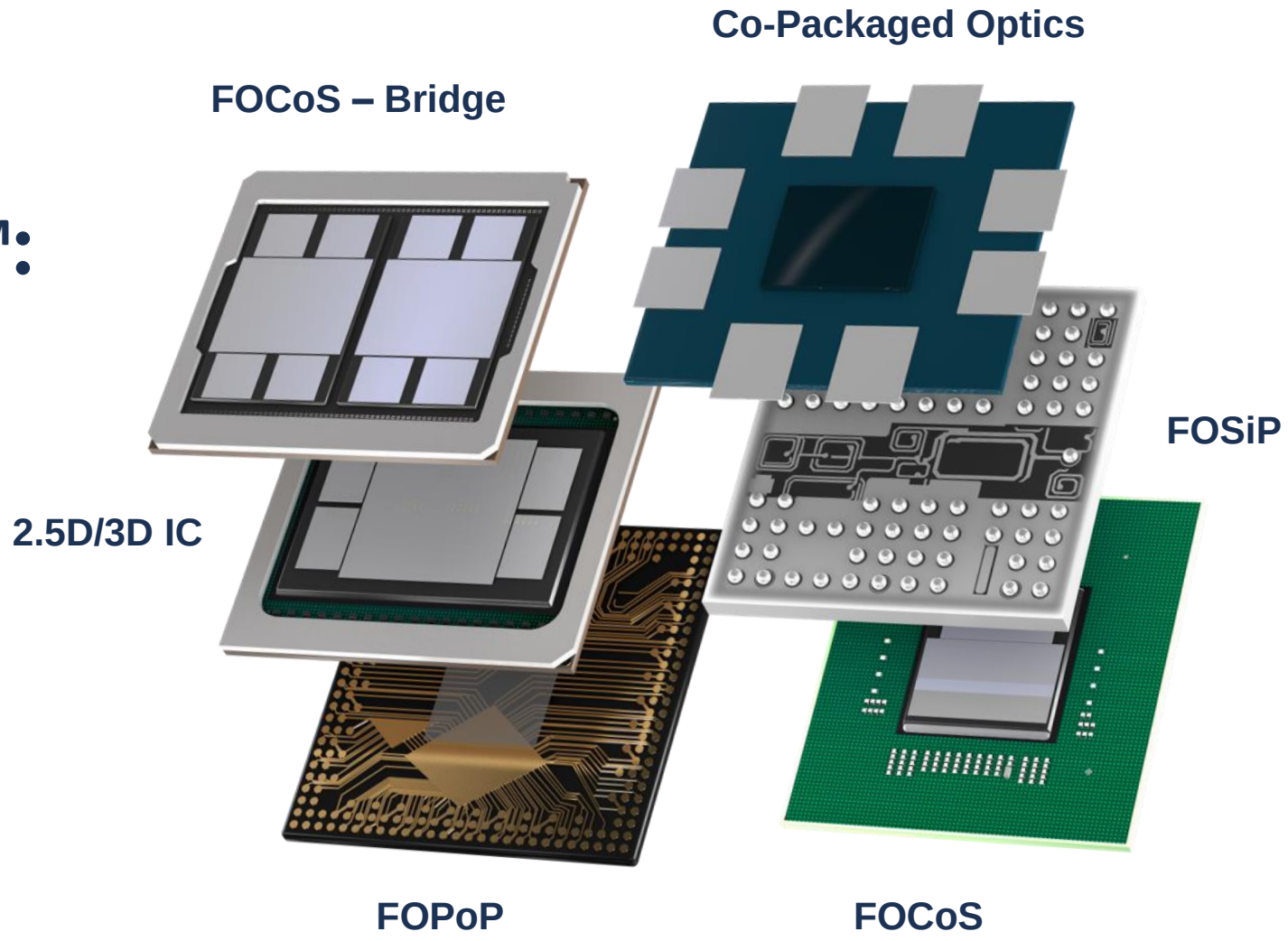
- CPU & GPU power demands rapidly increasing with GPU power expected > 1500W & CPU >600W by 2028
- Technical fluid temperature requires lower temp for effective heat dissipation while power increases
- Thermal management becomes the bottleneck for more efficient liquid cooling or lower temperature operations



Source: AMD

Leveraging VIPack™:

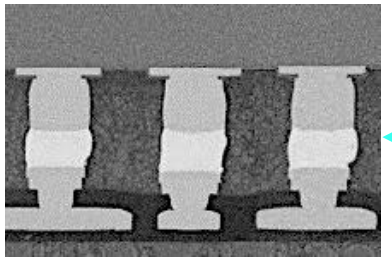
an advanced packaging platform designed to enable vertically integrated package solutions, featuring six technology pillars and underpinned with ASE's Integrated Design Ecosystem™ (IDE)



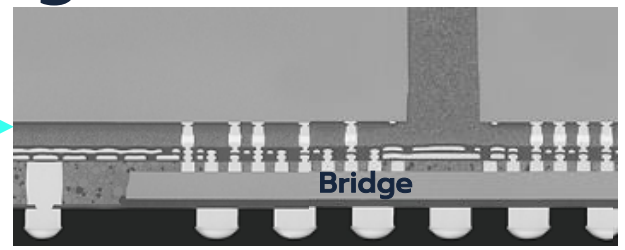
FOCoS and FOCoS-Bridge

FOCoS

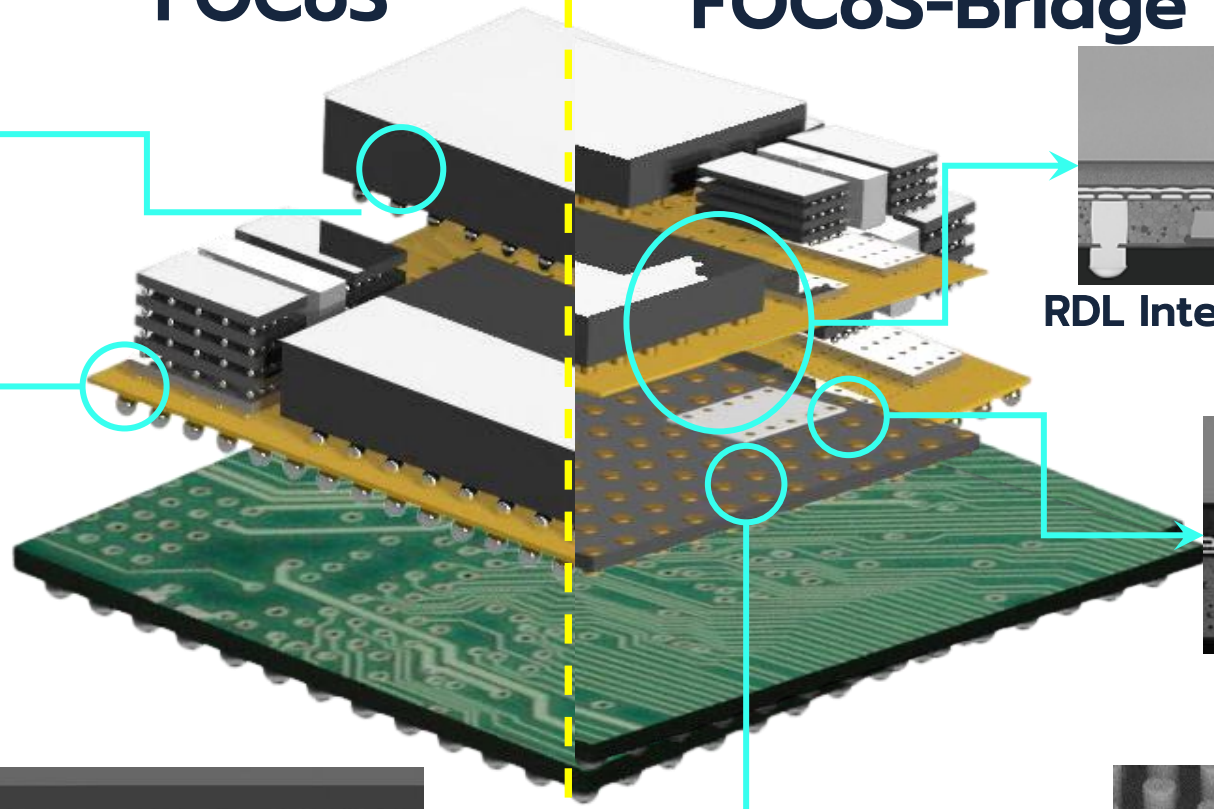
FOCoS-Bridge

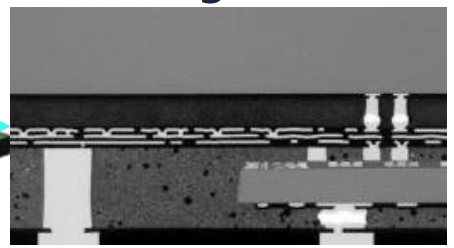


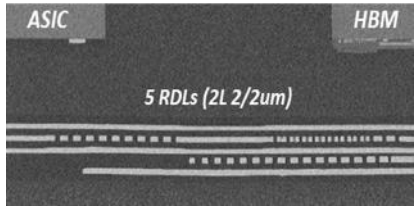
CoW uBump

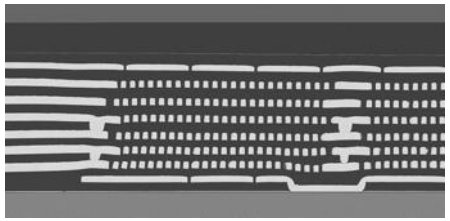


RDL Interposer with Embedded Bridge



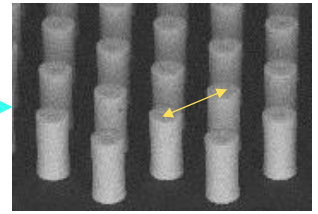




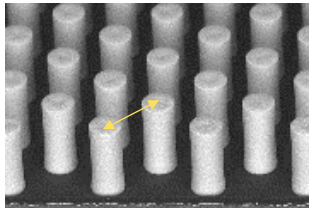


RDL Interposer

C4 bump pitch



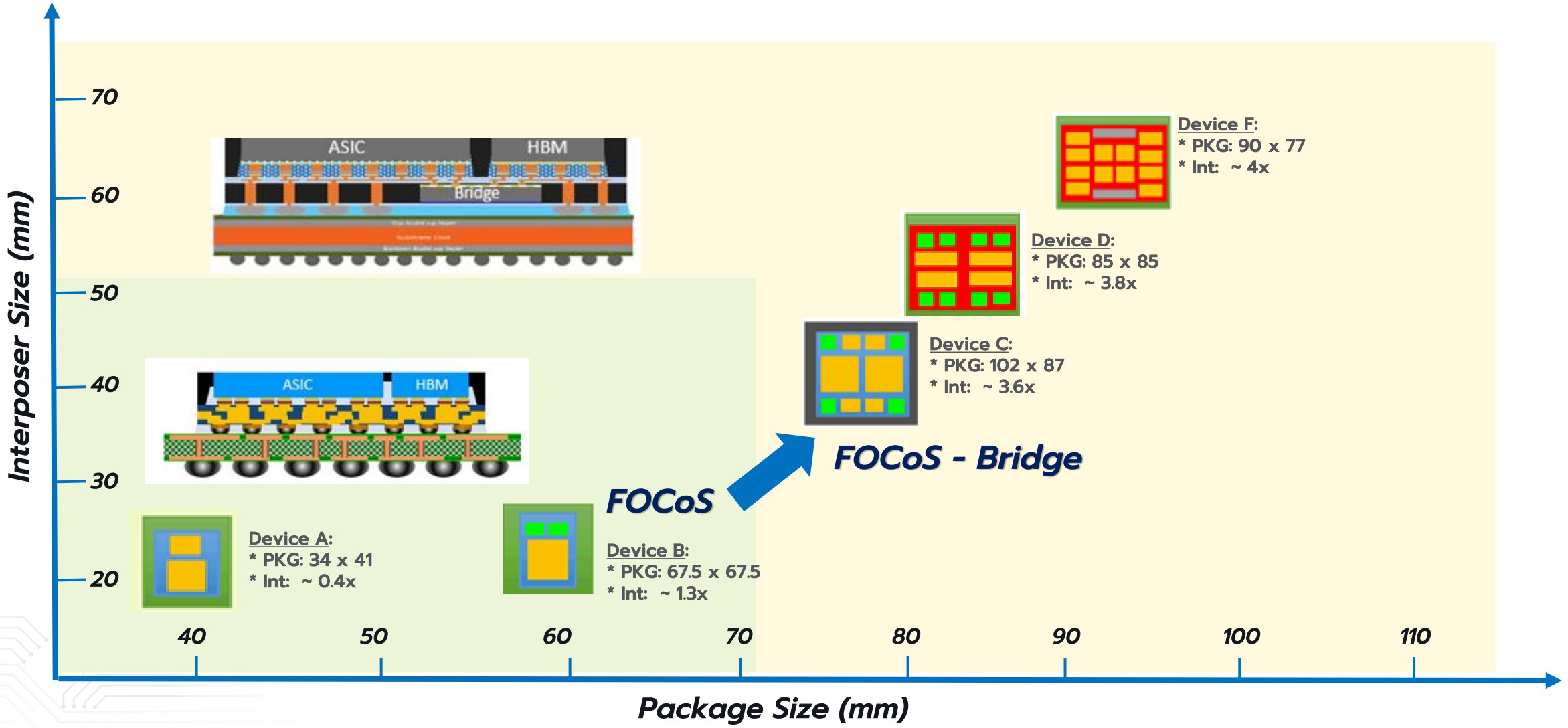
150um



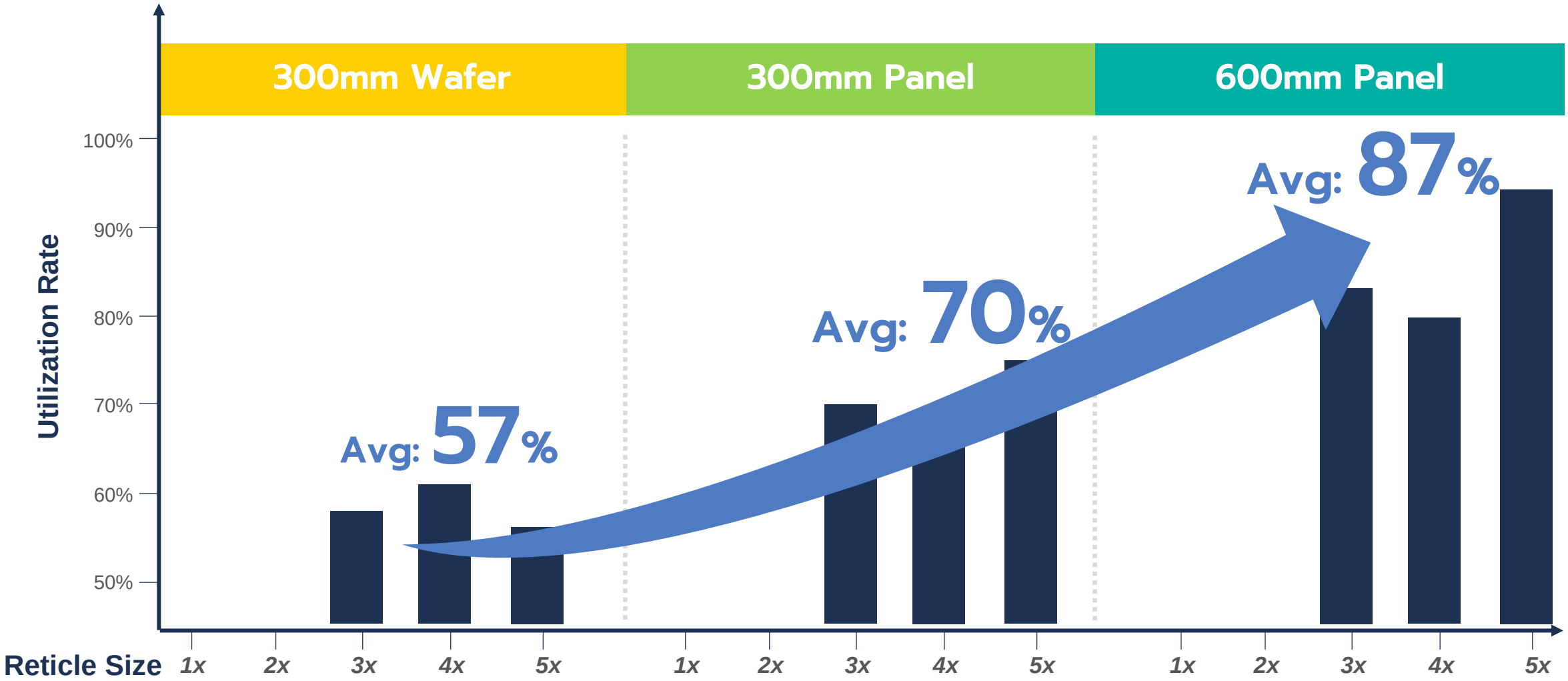
130um

FOCoS Extension Platform---FOCoS-B

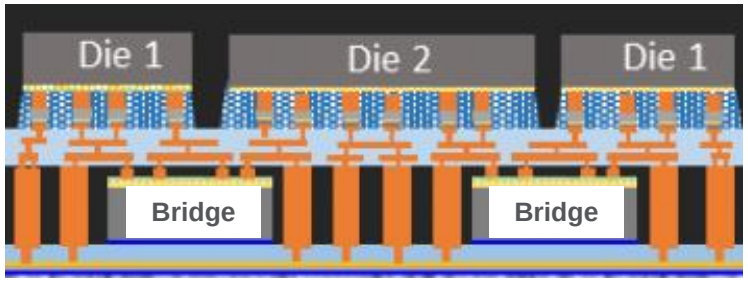
- Si bridge die provides high density D2D interconnect (< 1 um L/S) & enables embedded components – DTC, Power, Active, Passives for power delivery efficiency and performance improvement
- FOCoS-B provide larger scaling integration for larger reticle size (>3.3x) and larger package size (>70~150)

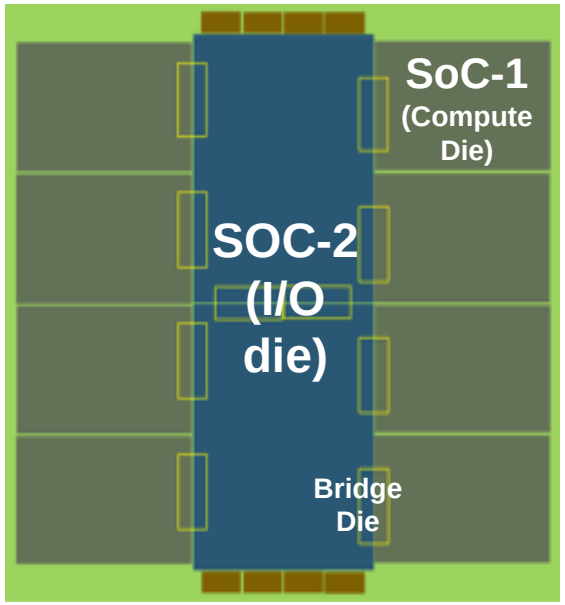


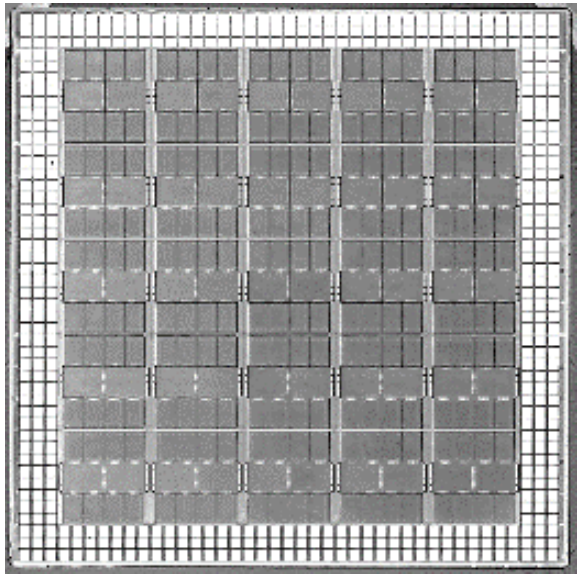
Panel Utilization vs. Reticle Size



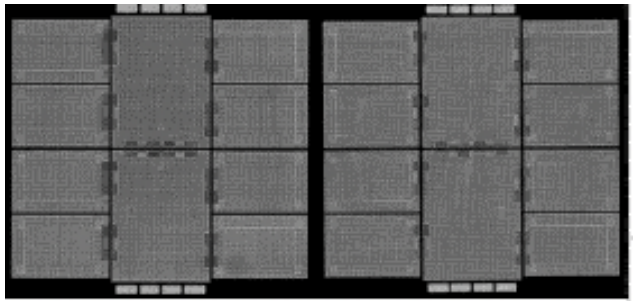
Panel FOCoS-Bridge



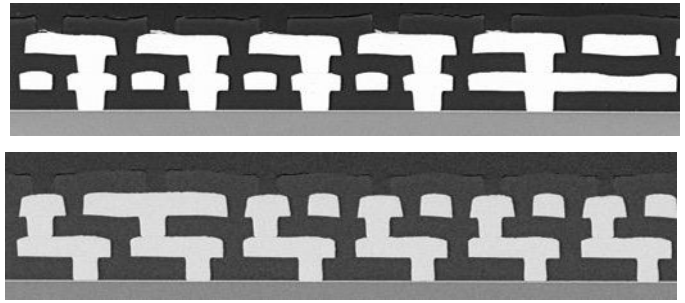




ubump pitch: 45um;
C4 bump pitch 150um

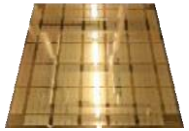
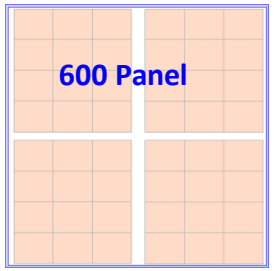
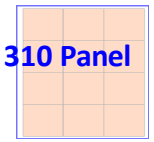


10 Chiplets w 10 silicon bridges



Fanout construction with stack & stagger vias by LDI

ASE Panel Roadmap

Platform	Application	PKG	Remark
 	 	Mobile	FO-MCM	• L/S=5/5 or above • FORDL as a Substrate
	HPC/AI	FOCoS-CL FOCoS-Bridge	• For L/S=2/2 or Less • Bridge and IPD

Panel

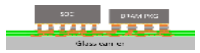
FO-MCM

600

600

300 (Compatible with 310)


RDL


FCB + DRAM


UBM


Ball Mount

FOCoS-CL
FOCoS-B

600

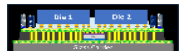
300 (Compatible with 310)

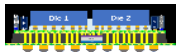
310

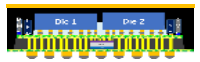

PI + Tall Cu pillar


D/A + MD + Grind


RDL + uPad

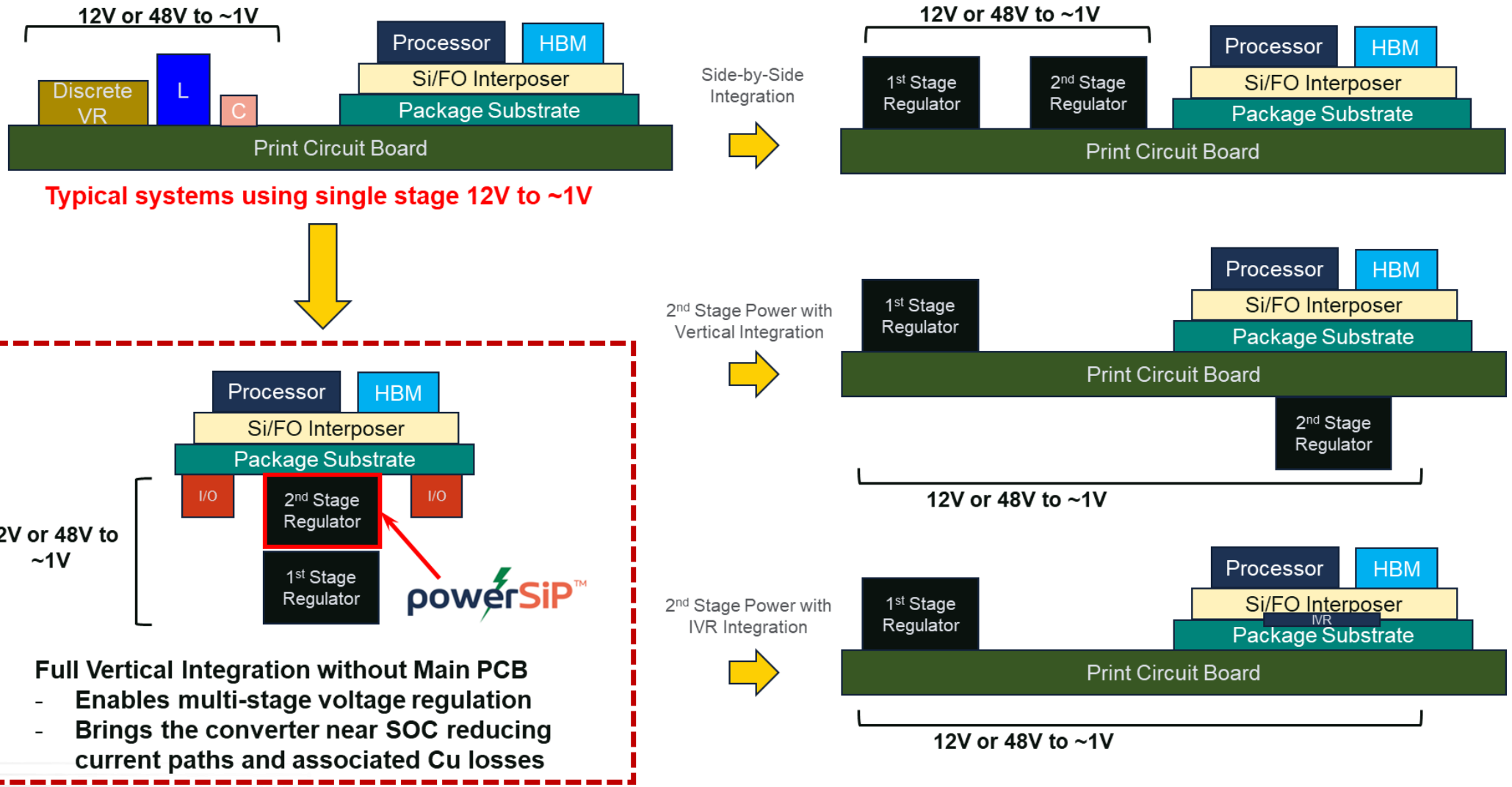

FC + CUF + MD


C4

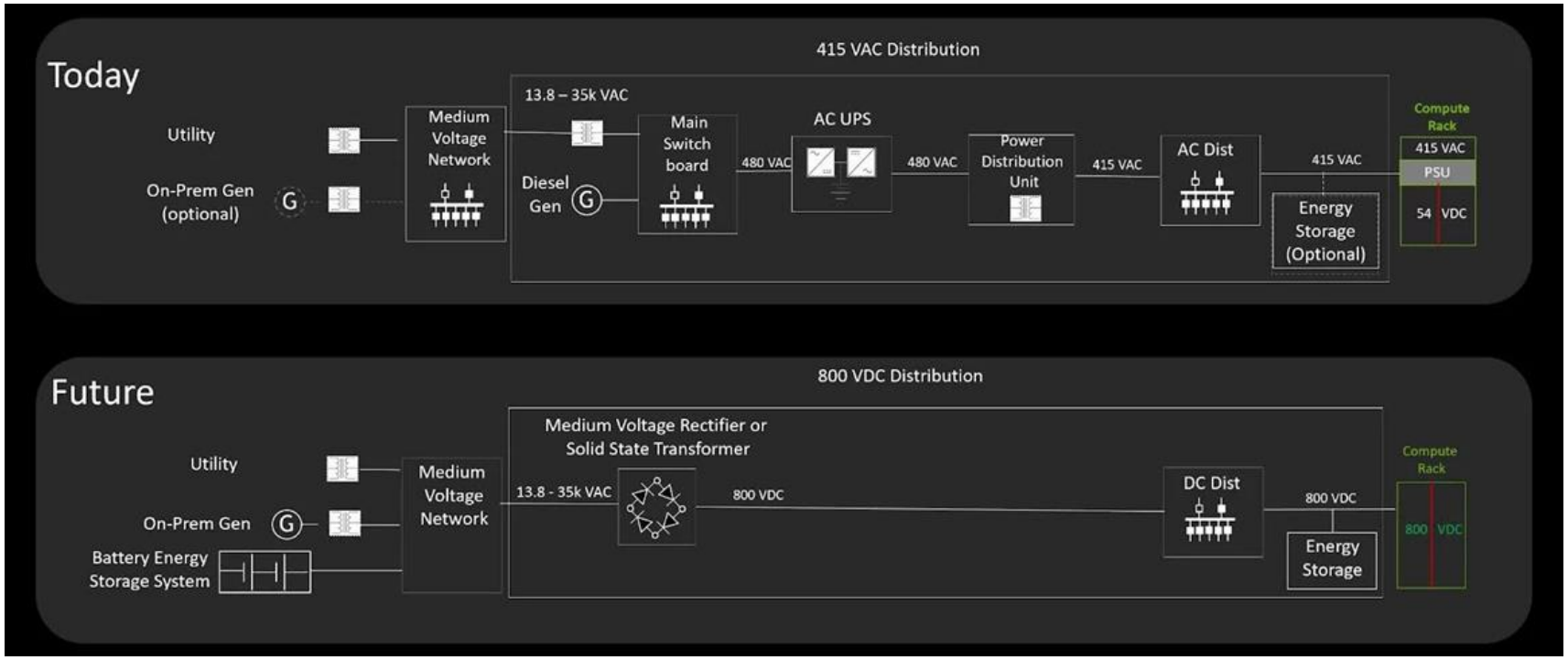

CoS

20

Power Integration for HPC Systems



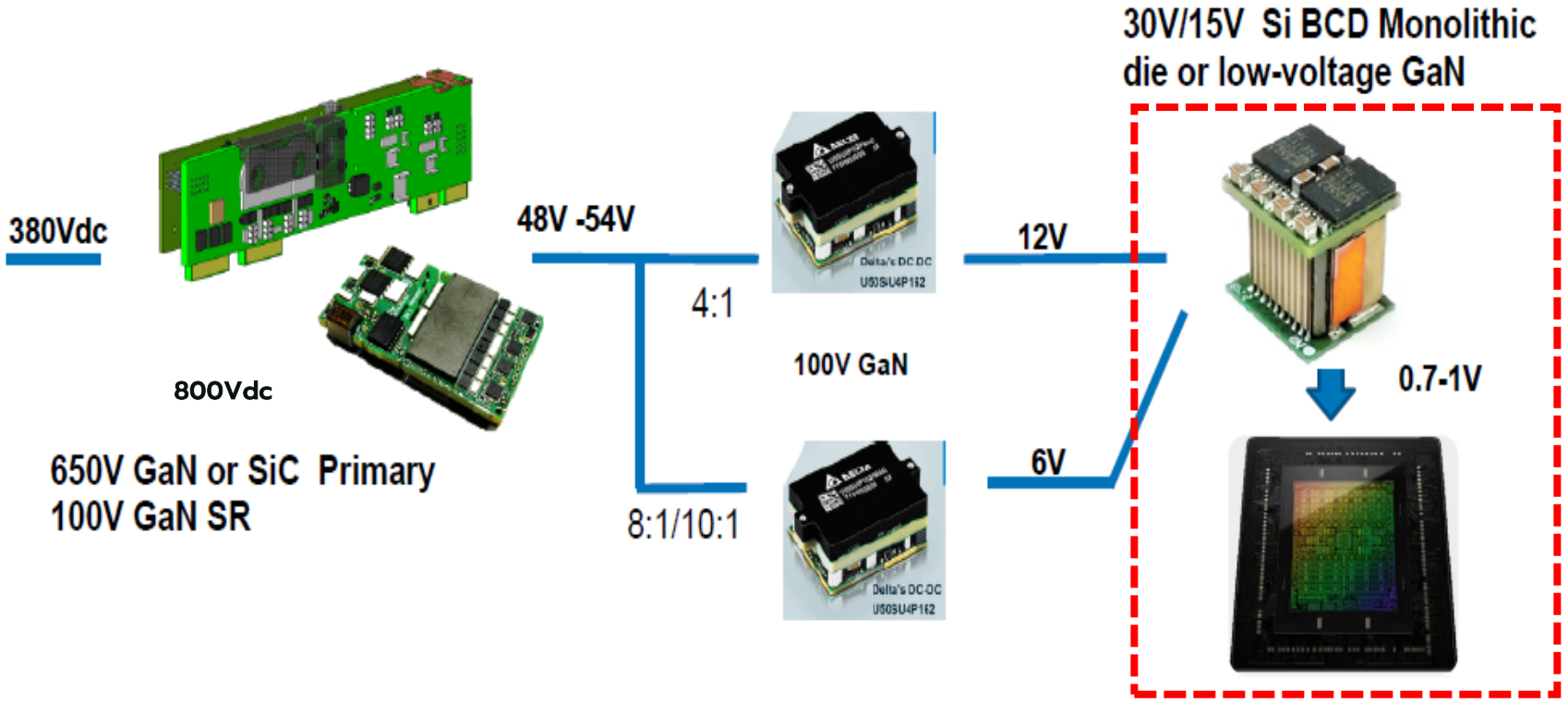
Migration to 800 HVDC Systems



Ref: Nvidia

- Transitioning to 800 VDC power distribution coupled energy storage integration
- Simpler DC distribution system with fewer points of failure, better overall system reliability
- Aligns with EV and renewable industry

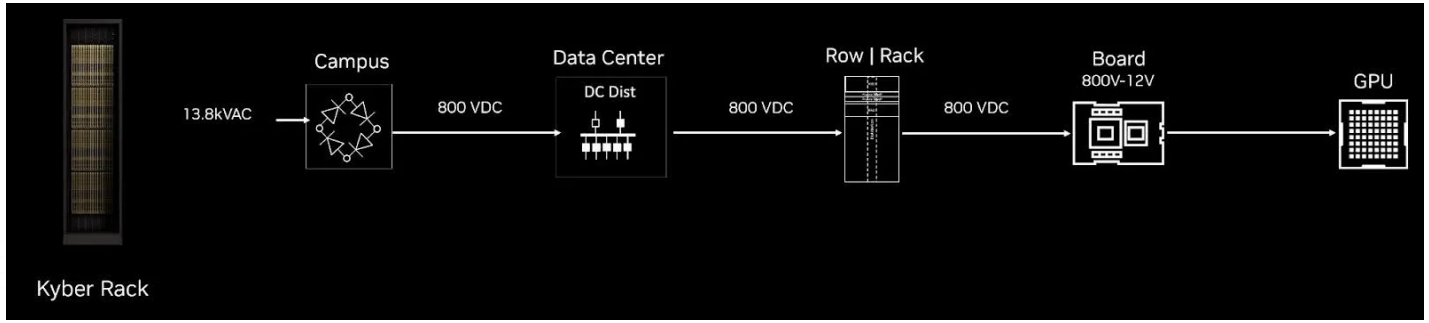
High Voltage DC to GPU Supply Voltages



GaN devices have superior performance (higher at 100V, 650V); cost still two times higher than Si

BCD Bipolar-CMOS-DMOS

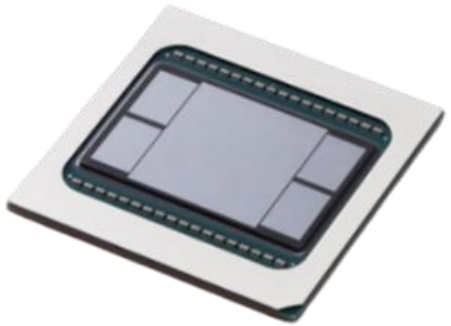
Cyntec, APEC 2025



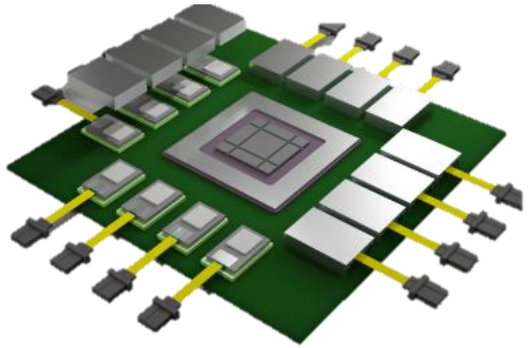
Ref: Nvidia

Heterogeneous Integration: Silicon Photonics

Chiplets



CPO
(Co-Packaged Optics)



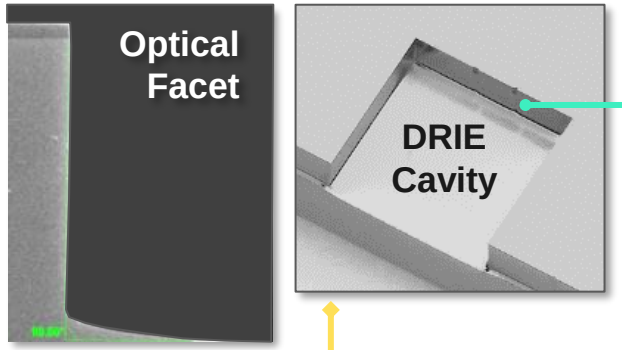
Data Center



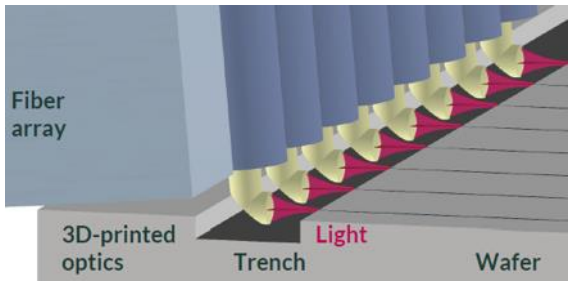


SiPh Optical Engine (OE) Toolbox for Optical Bandwidth

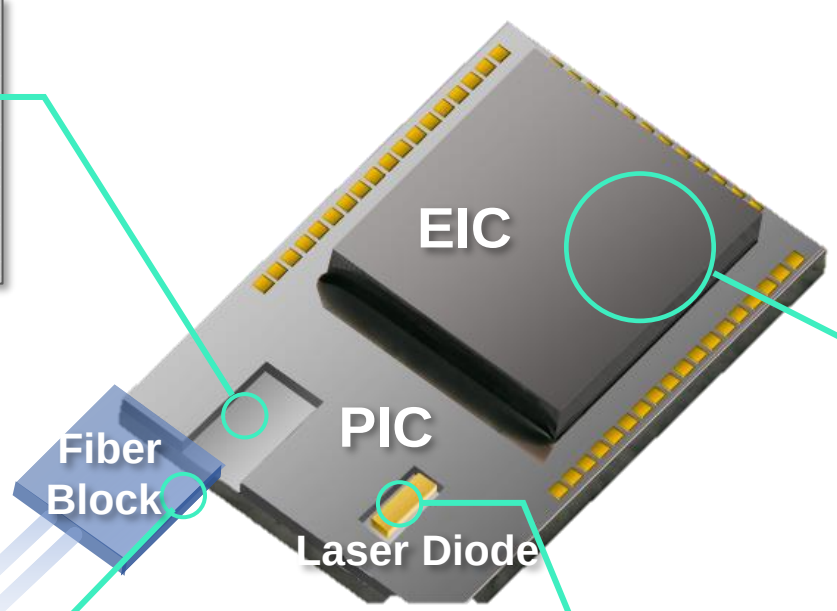
Optical Coupling



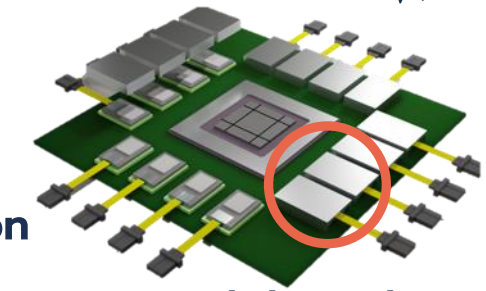
Enable 5x UPH



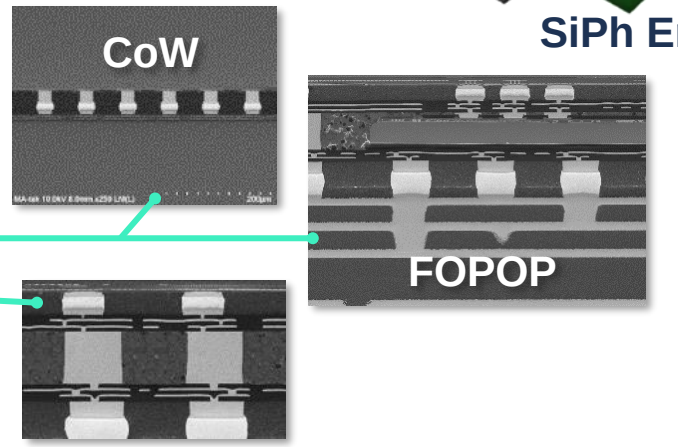
Fiber Passive Alignment

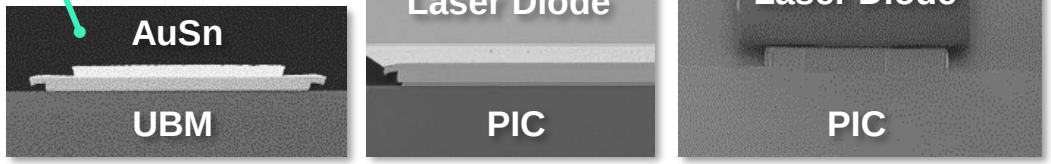


3D EIC/PIC Integration



SiPh Engine



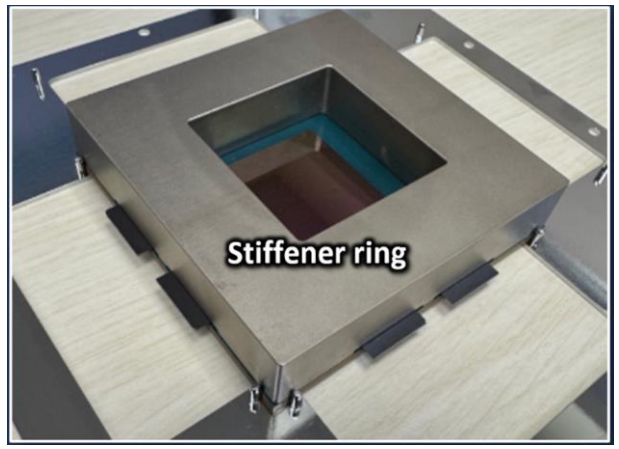


Laser Diode Integration ⟷ Sub-µm Accuracy

OE & ASIC on Substrate Challenges

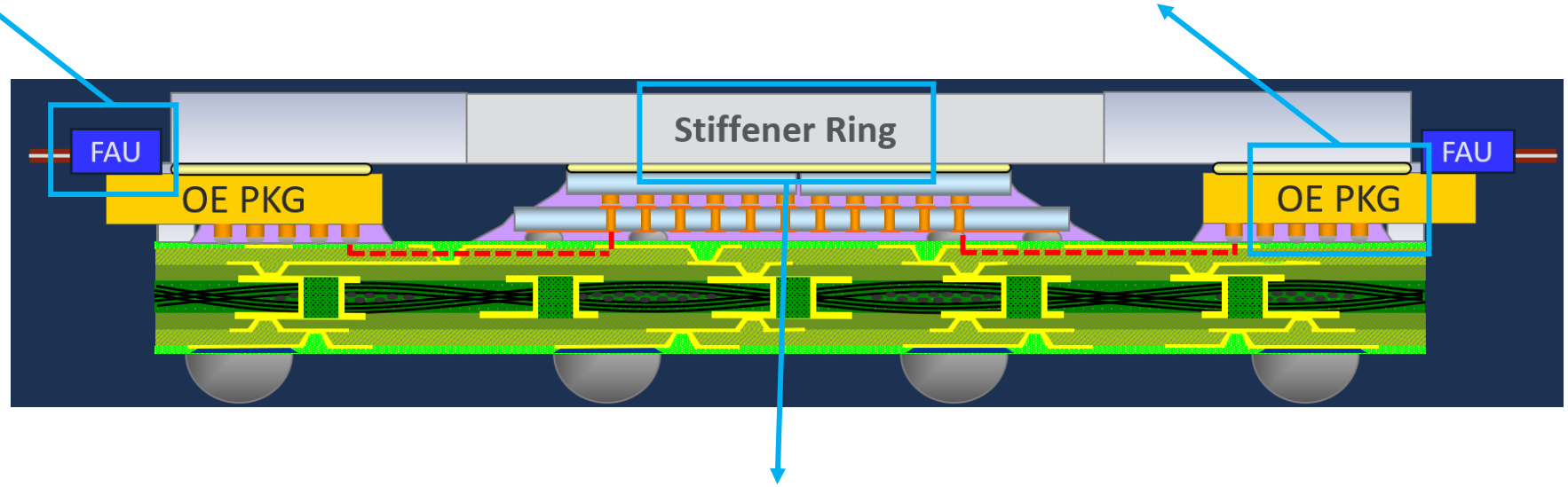
Challenge 1: Fiber Attach

- Coupling approach (PA/AA)
- FAU Detachability and reflow compatibility



Challenge 2: Optical Engine

- Engine size & EIC/PIC integration approach & warpage control
- OE test solution (KGOE) & ensure optical facet quality

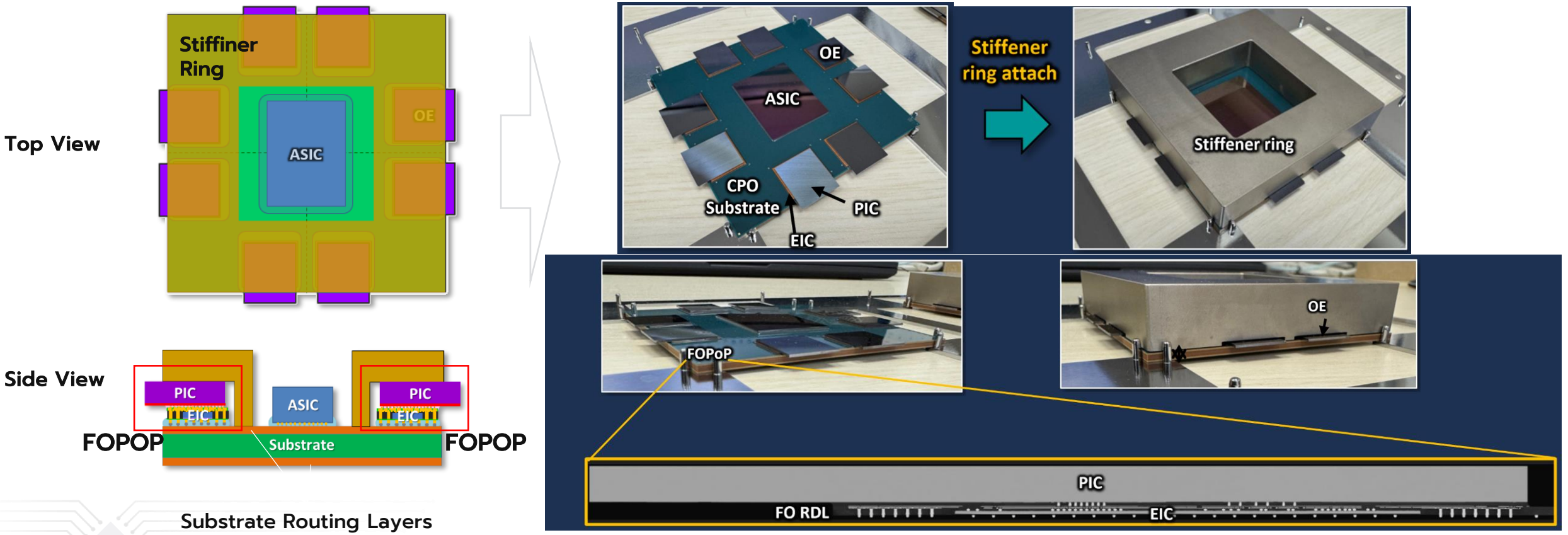


Challenge 3: On Substrate

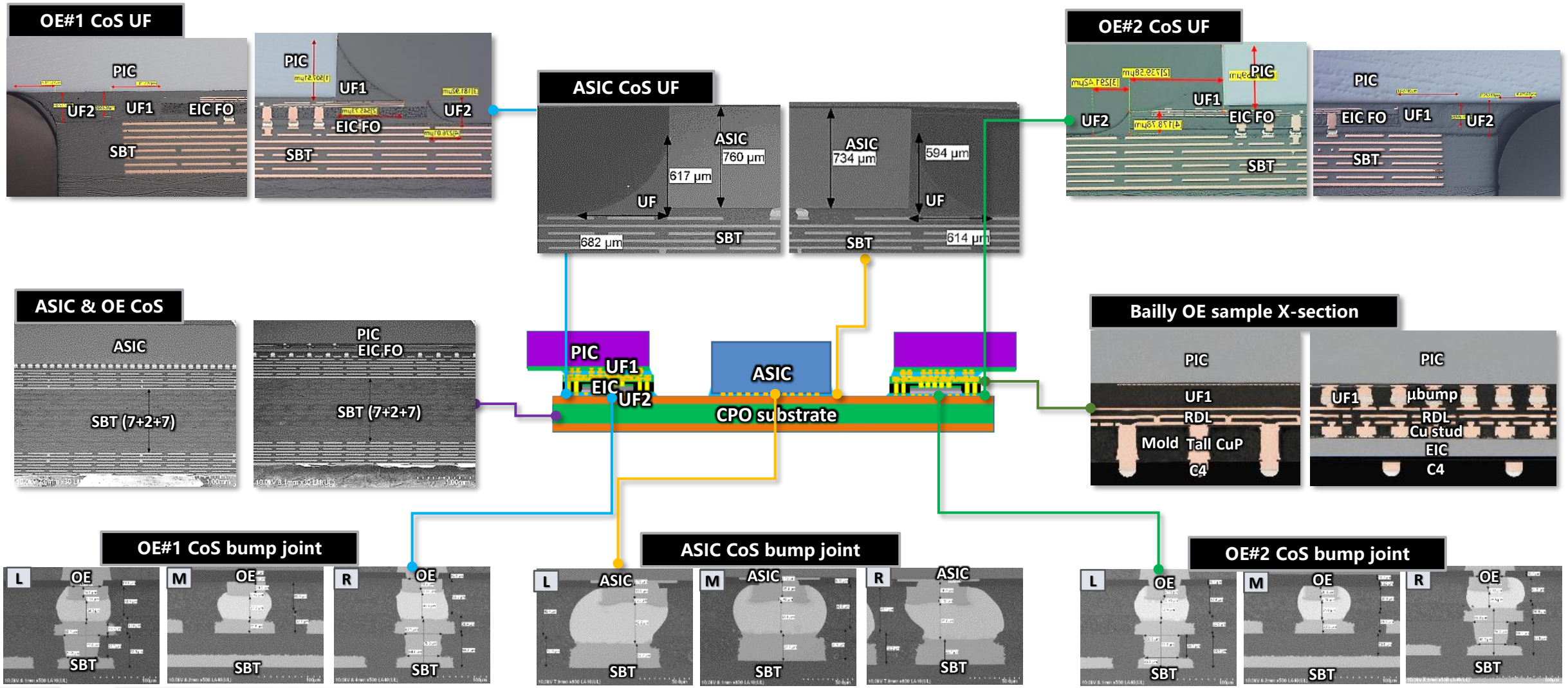
- Overall warpage control
- Thermal dissipation

CPO Test Vehicle Demonstration (OE on Substrate Process)

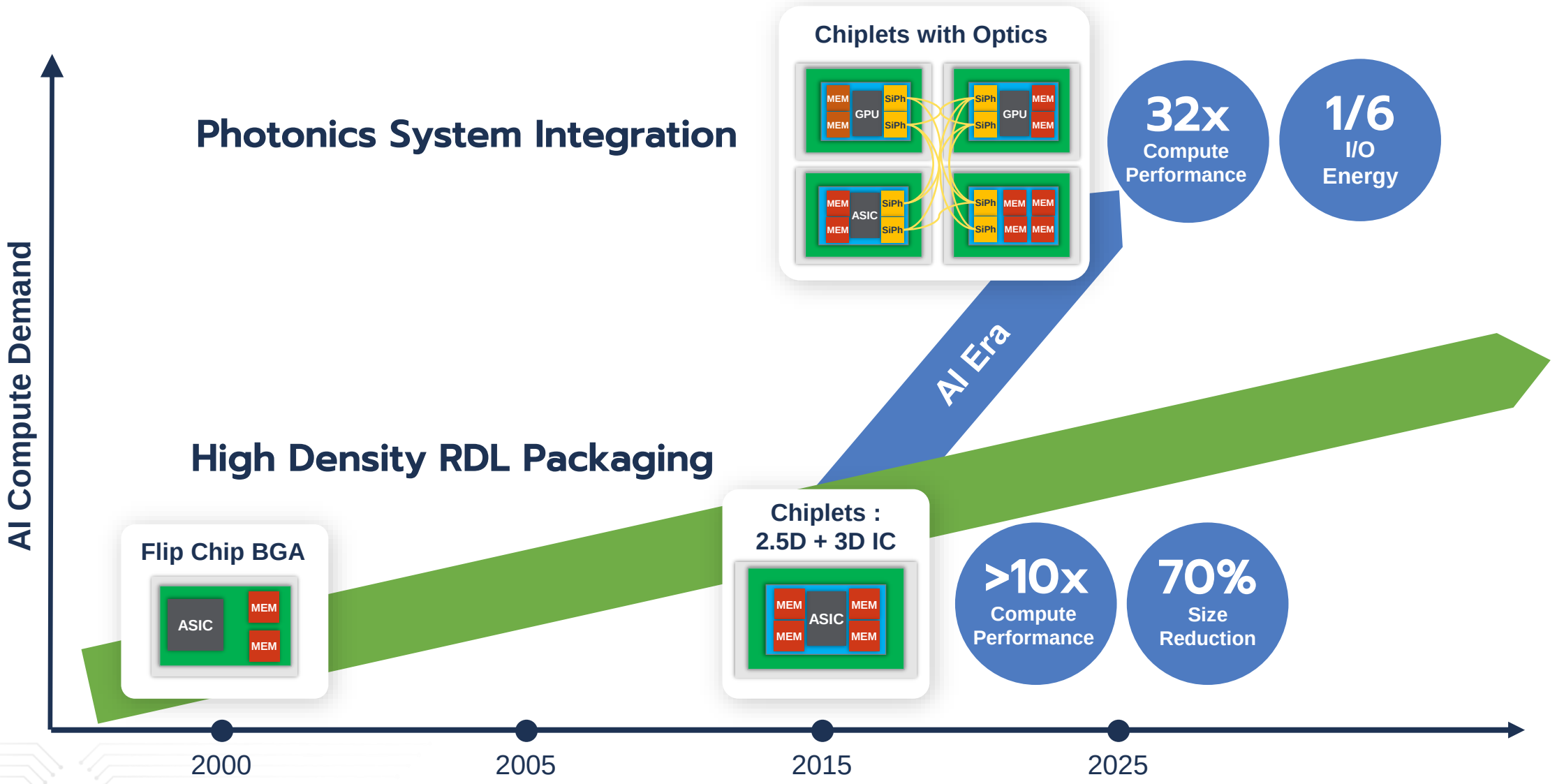
- CPO substrate & stiffener ring & ASIC/EIC designed and manufactured by ASE
- ASIC & 8x OE FCB on substrate with one time reflow process; **>75 mm x 75 mm package size**
- Good bump joint by X-ray and cross section check



Co-Packaged Optic Solutions

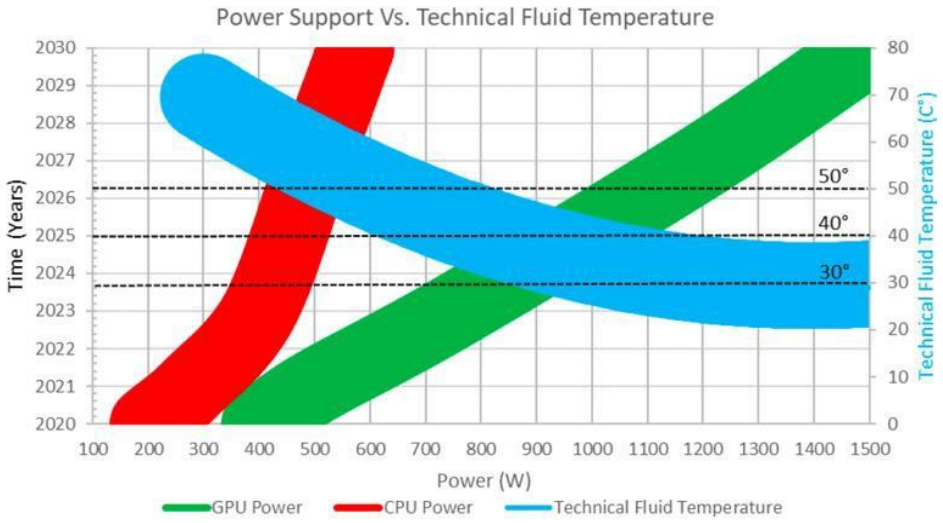


Packaging Evolution for AI Systems



Advanced Thermal Solutions for AI

Rising Power Demands Drive Next-Gen Cooling Requirements

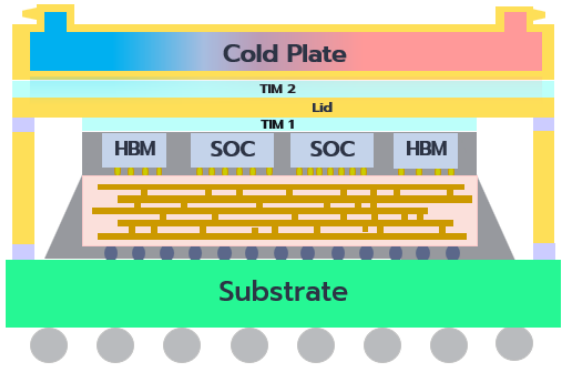


Source:AMD

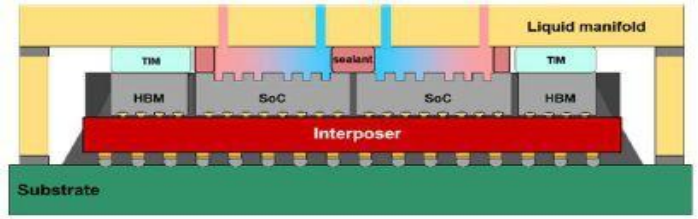
- AI/compute power rapidly increasing:
  - **GPU power > 1500W**
  - **CPU power > 600W (expected by 2028)**
- Thermal fluid temperature must decrease to effectively dissipate increasing heat
- **Thermal management is now a critical bottleneck** for enabling efficient liquid cooling and maintaining lower operational temperatures

Current vs. Next-Gen Package Level Thermal Solutions

System Solution



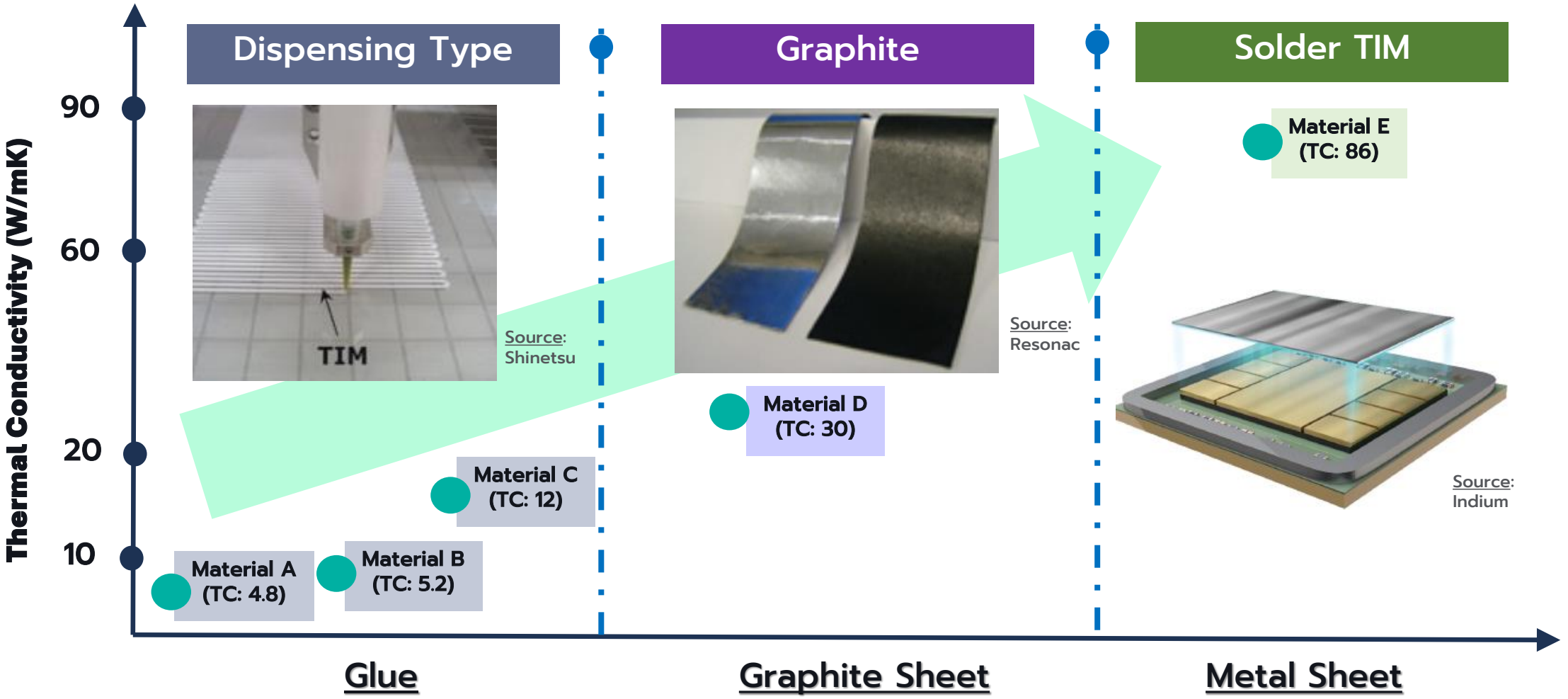
Silicon Solution



TSMC

- **Current:** Uses Thermal Interface Material (TIM) between die/lid and cold plate, but interface resistance degrades performance
- **Next:** Direct liquid-cooled solutions eliminate thermal interfaces, improving dissipation and supporting higher chip power densities

ASE Thermal TIM Solutions



Advanced Packaging Innovation to Meet AI Challenges

= Demand = Solution

Packaging Architecture & Technology

Performance	Memory	Area
2x – 7x increased compute workload	2x - 10x network bandwidth & memory capacity	2x - 12x reticle size 2x-6x PKG size (~120x180 mm²)

- ✓ Chiplets & Heterogeneous Integration (CPU, GPU, APU, XPU, HBM, AI accelerator, IPD, DTC)
- ✓ 2.5D Si TSV, FOCoS-CF/CL, FOCoS-B, SiPh, Panel FOCoS

Packaging Infrastructure & Ecosystem

Power	Thermal
Exascale 21 MW to zettascale 500 MW power delivery	~3 kW thermal coolings

Power	Thermal
✓ PowerSiP (VRM, IVR Integration) ✓ Backside power ✓ CPO	✓ Cooling structure, liquid cooling, Si microchannel, new TIM materials

ASE Confidential Proprietary

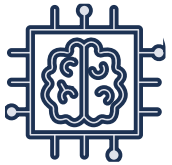
Summary



- **AI and Data** continue to fuel semiconductor innovation, with exponential proliferation through 2030 and beyond that **will shape global life and lifestyle in unimaginable ways.**



- Collectively, our industry is accelerating the AI economy through **Heterogeneous Integration advancements.**



- Packaging creativity is enabling seamless integration of multiple chiplets, SiPs, modules into one single package optimized for **enhanced functionality and operating characteristics.**

Thank you